

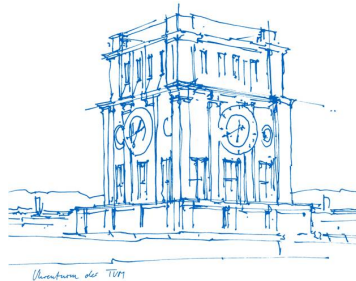
HeatPath–GNN

Learning where to place conductive particles

Thermal transport through
particle contact networks

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Motivation: conductive material is a limited resource

Heat transfer through particle beds

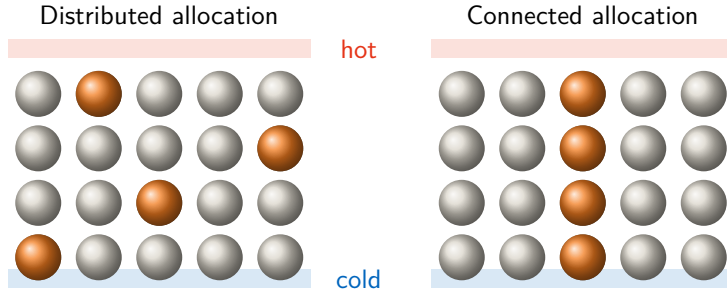
- Heating and cooling granular products
- Thermal storage and temperature control
- Transport depends on contacts between particles

A design choice

- Highly conductive material may be costly
- The material budget is fixed
- Its position within the bed can change the conducting network

Can we obtain better heat transfer from the same amount of conductive material?

The allocation problem



Orange: high conductivity. Grey: low conductivity. Schematic only.

Geometry and material amount stay fixed. Only particle labels change.

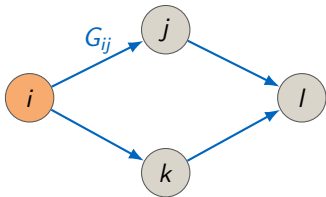
Thirty packings with the same nominal porosity

Quantity	Value
Particles	500
Diameters [mm]	1.5 / 2.0 / 2.5
Counts	100 / 300 / 100
Nominal porosity	0.38
Independent seeds	30

- Periodic boundaries in x and y
- Two fixed thermal walls in z
- No gravity during generation
- Grow small spheres, then relax
- Accept final translational $KE \leq 10^{-12}$ J

DEM: Hertz–Mindlin, $E = 50$ MPa, $\nu = 0.25$, restitution 0.30, friction 0.50.

A contact network represents steady heat conduction



Nodes: particles Edges: contacts

Conservative balance at each particle

$$\sum_j G_{ij}(T_i - T_j) + \sum_w G_{iw}(T_i - T_w) = 0$$

$$k_{\text{eff}} = \frac{QH}{A(T_{\text{hot}} - T_{\text{cold}})}$$

Q : wall heat rate, H : bed height, A : wall area.

Solve particle temperatures, then calculate the bed conductivity.

Thermal conductance follows contact size and material

Particle–particle contact

$$a_{ij} = \sqrt{R_{ij}^* \delta_{ij}}, \quad R_{ij}^* = \frac{R_i R_j}{R_i + R_j}$$

$$G_{ij} = \frac{4a_{ij}k_i k_j}{k_i + k_j}$$

δ_{ij} : geometric overlap.

a_{ij} : Hertz contact radius.

Ideal circular constriction model. Its small-contact assumption needs physical assessment.

Material assignment

$$k_i \in \{1, 10\} \text{ W m}^{-1}\text{K}^{-1}$$

Isothermal wall

$$G_{iw} = 4a_{iw}k_i$$

Two spreading resistances in series give the particle-contact formula.

Every method uses the same conductive-material budget

Diameter [mm]	1.5	2.0	2.5
Total particles	100	300	100
High-conductivity particles	10	30	10

$$\max_{\mathbf{s}} k_{\text{eff}}(\mathbf{s}), \quad s_i \in \{0, 1\}, \quad \sum_{i \in \mathcal{B}_b} s_i = q_b$$

$s_i = 1$: conductive particle. $\mathcal{B}_b$: size class. q_b : class quota.

Fifty of 500 particles are conductive, with exactly 10% in each size class.

Reference allocations and the search teacher

Simple reference methods

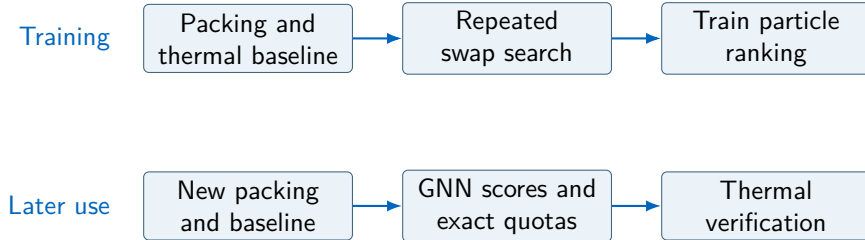
- Random: mean of 100 feasible allocations
- Degree: favour particles with many contacts
- Heat path: rank homogeneous-baseline heat throughput

Same-size swap search

- Exchange one high and one low label
- Both particles belong to the same size class
- Accept conductivity improvements
- Five restarts, 3000 proposals each

The teacher is the best allocation found within a fixed search budget.

Training and later use have different workflows



The learned ranking replaces repeated candidate search at prediction time.

The GNN combines local geometry with baseline physics

Eight particle features

- Diameter, height and contact degree
- Hot-wall, cold-wall and anchored flags
- Homogeneous temperature
- Homogeneous heat throughput

Six contact features

- Contact radius
- Relative displacement in x, y, z
- Absolute vertical alignment
- Homogeneous absolute contact heat rate

Three message-passing layers, width 64. The MLP receives the same particle features, without neighbour messages.

Scores become a feasible allocation

Supervised ranking

- Search restart selections provide soft particle targets
- Listwise ranking within each size class
- Additional binary cross-entropy term
- Validation controls early stopping

Selection at prediction time

- Train three repeats per model
- Average within-class ranks
- Select the top 10 / 30 / 10 particles
- Verify conductivity with the thermal solver

Each test packing is withheld from model fitting

28 training packings
weights and scalers

1 validation packing
early stopping

1 test packing
reported allocation

- Repeat for all 30 test packings
- Up to 400 epochs and patience 60
- Compare methods on the same held-out geometries

This tests transfer between realizations of the specified packing.

Gain and recovery answer different questions

$$\text{Gain}_m = 100 \frac{k_m - k_{\text{random}}}{k_{\text{random}}}$$

$$\text{Recovery}_m = 100 \frac{k_m - k_{\text{random}}}{k_{\text{search}} - k_{\text{random}}}$$

Gain

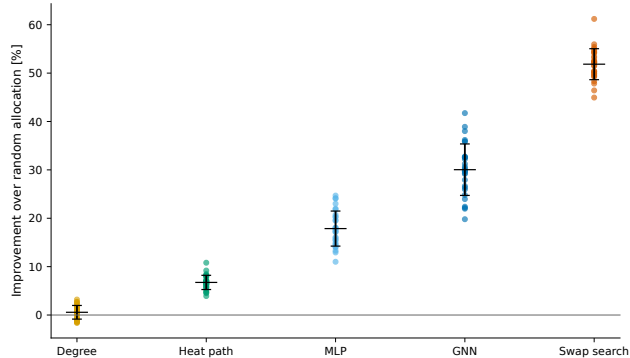
How much better is this allocation than random?

Recovery

How much of the search improvement does the model obtain?

Compute both per packing, then average. Error bars describe sample SD across packings.

GNN improves conductivity by 30% over random

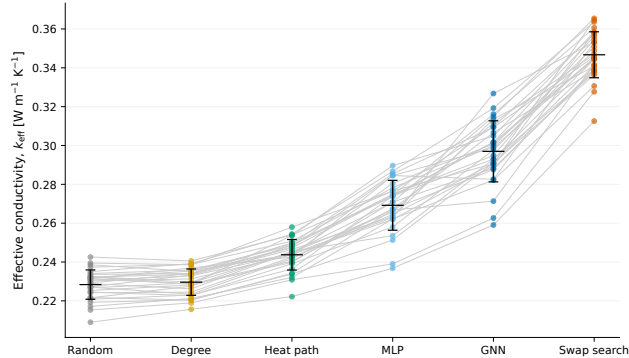


Mean improvement

Degree	0.6%
Heat path	6.7%
MLP	17.9%
GNN	30.0%
Search	51.8%

Same geometry and material budget for every comparison.

Absolute conductivity remains below the search reference

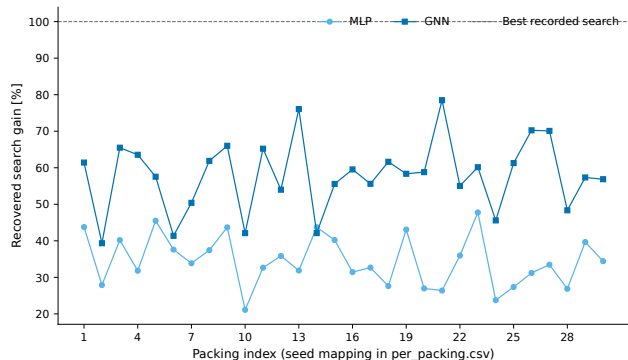


Mean k_{eff} [$\text{W m}^{-1} \text{K}^{-1}$]

Random	0.2284
MLP	0.2692
GNN	0.2970
Search	0.3467

Grey lines connect results on the same packing.

GNN beats MLP on 29 of 30 held-out packings



Recovered search improvement

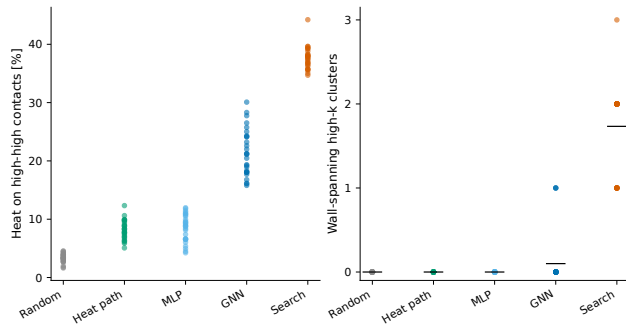
GNN: **58.0%**

SD: 9.8 percentage points

MLP: 34.5% mean

Improvement is consistent, but performance varies between packings.

Contact-network diagnostics help interpret allocation

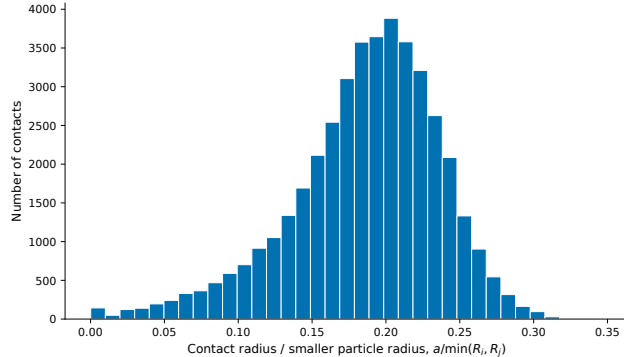


Two diagnostics

- Fraction of absolute contact heat carried by high-high contacts
- Number of high-conductivity clusters spanning both walls

These describe the selected network. They do not prove causality.

Numerical checks pass, but contact sizes need scrutiny



Recorded checks

- Maximum final KE:
 $8.4 \times 10^{-13} \text{ J}$
- Maximum energy imbalance:
 9.01×10^{-12}

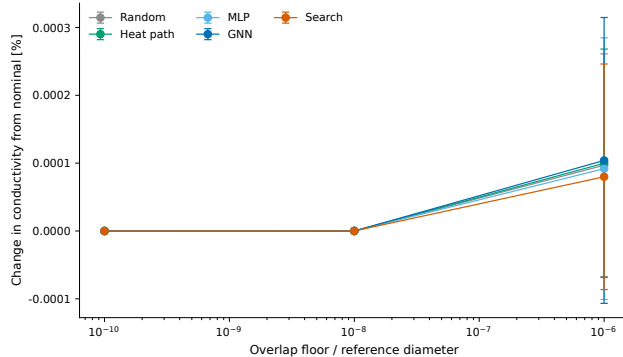
Contact radius / smaller radius

Median: 0.192

95th percentile: 0.258

Maximum: 0.347

The numerical overlap floor has little influence



Fixed allocations

- Vary the numerical overlap floor
- Largest absolute conductivity change: 0.000923%
- Sensitivity is negligible over the tested range

This does not test stiffness, loading or alternative contact laws.

Fast scoring still requires two thermal solves

Recorded computational work

- Search: about 14,964 thermal solves per packing
- GNN ensemble scoring: 8.63 ms mean
- MLP ensemble scoring: 0.295 ms mean

Complete prediction procedure

- Homogeneous thermal baseline
- Feature construction and model scoring
- Quota selection
- Thermal verification

The campaign demonstrates fewer candidate evaluations, not a measured end-to-end speedup.

Conclusions and next research steps

What the campaign establishes

- Material placement strongly affects modelled conductivity
- GNN gains 30% over random allocation
- Neighbour information improves on the MLP
- 58% of best-found search gain is recovered

What remains to establish

- Physical sensitivity of contact areas
- Transfer to other packing specifications
- GNN-initialized search versus random starts
- Complete runtime and manufacturing feasibility

A learned particle ranking captures useful thermal-network structure.